

Question_1:

Code:

```
1  /*
2  Question No: 1
3  Given n values, construct a Binary Search Tree by inserting
4  the values in the given order. Print the preorder traversal.
5  Preorder traversal:
6  Root -> Left -> Right
7  */
8
9  #include <iostream>
10 using namespace std;
11
12 struct node {
13     int data;
14     struct node *left,*right;
15 };
16
17 struct node* newNode(int x) {
18     node* t=new node;
19     t->data=x;
20     t->left=t->right=NULL;
21     return t;
22 }
23 node* insert(node* root,int x) {
24     if(root==NULL) return newNode(x);
25     if(x<root->data)
26         root->left=insert(root->left,x);
27     else
28         root->right=insert(root->right,x);
29     return root;
30 }
31 void preorder(struct node* root) {
32     if(root) {
33         cout<<root->data<<" ";
34         preorder(root->left);
35         preorder(root->right);
36     }
37 }
```

```
38  int main() {
39      int n,x;
40      cin>>n;
41
42      node* root=NULL;
43      for(int i=0;i<n;i++) {
44          cin>>x;
45          root=insert(root,x);
46      }
47      preorder(root);
48  }
```

Output:

Test Case_1:

```
7
200 120 80 160 280 240 320
200 120 80 160 280 240 320
PS C:\Users\Prabin Yadav\Documents\III SE
```

Test Case_2:

```
9
34 67 10 90 410 810 40 20 60
34 10 20 67 40 60 90 410 810
PS C:\Users\Prabin Yadav\Documents\III
```